



CCA GAME DEVELOPMENT

Listening for Triggers

Module 2 — Session 6

Teacher: Annisa Hendrawan

Junior High School — Global Prestasi School



`KEYDOWN` → `K_RIGHT`

`event.type == KEYDOWN`

Today's Goal

By the end of this session:



Understand what a game event is

01



Listen for keyboard input

02



Use `pygame.event.get()`

03



Use `K_LEFT` and `K_RIGHT` to control the player

04

How does a game know what we do?

A game can listen for events. An event can be:



A key is pressed



The player closes the window



The mouse is clicked

INPUT



GAME DETECTS



GAME RESPONDS

Example: Press RIGHT → Player moves right

What is an Event?

An event is something that happens while the game is running.

“Did the player do something?”



KEY PRESS

A keyboard key is pressed



MOUSE CLICK

The mouse is clicked



QUIT

The player closes the game

The game needs to listen for these events.

How does Pygame listen?

Pygame checks for events using:



```
pygame.event.get()
```

This gives us the events that happened.

Then the game can decide what to do.



CHECK

“What happened?”

Keyboard Triggers

We can check if a keyboard key was pressed.



```
if event.type == pygame.KEYDOWN:
```

KEYDOWN

KEYDOWN means:

A key was pressed.

Then we can check which key was pressed.

Trigger: RIGHT

Which key should move the player right?



```
pygame.K_RIGHT
```

Example:



```
if event.key == pygame.K_RIGHT:  
    player_x += 20
```



K_RIGHT



RIGHT key → X increases → Player moves right

Trigger: LEFT

Which key should move the player left?



```
pygame.K_LEFT
```

Example:



```
if event.key == pygame.K_LEFT:  
    player_x -= 20
```



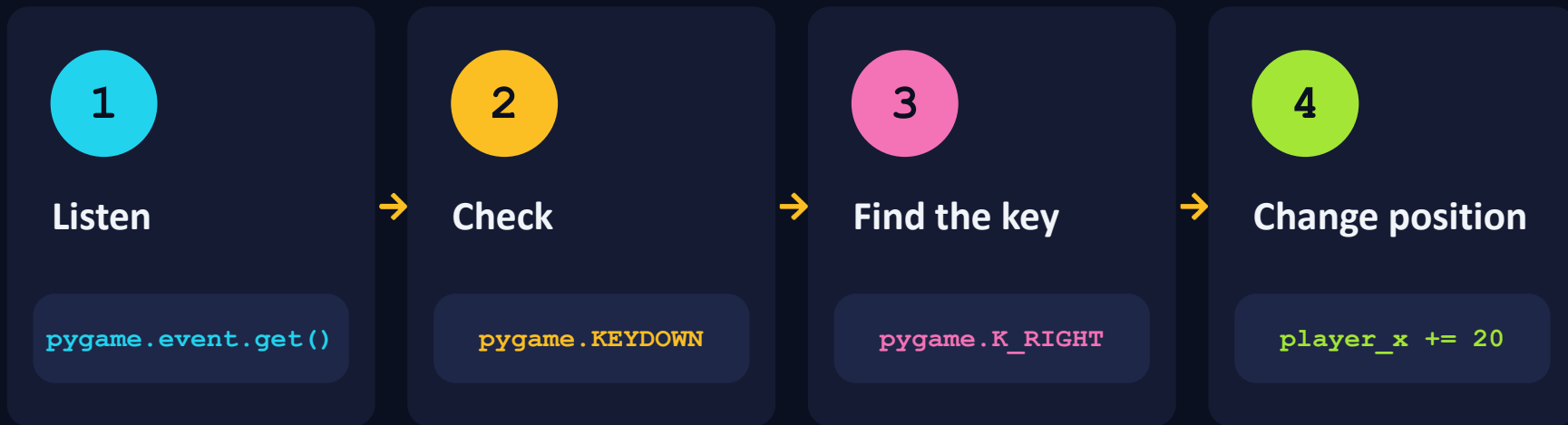
K_LEFT



LEFT key → X decreases → Player moves left

From Key Press to Movement

Follow the steps:



KEY → TRIGGER → ACTION

Let's Code!



Open the Session 6 starter code



Find the event loop



Add the keyboard trigger



```
for event in pygame.event.get():  
    if event.type == pygame.KEYDOWN:  
        if event.key == pygame.K_RIGHT:  
            player_x += 20  
        if event.key == pygame.K_LEFT:  
            player_x -= 20
```

Run → Press LEFT / RIGHT → Observe

What does each line do?

```
for event in pygame.event.get():
```



Listen for events

```
if event.type == pygame.KEYDOWN:
```



Check if a key was pressed

```
if event.key == pygame.K_RIGHT:
```



Check if it was the RIGHT key

```
player_x += 20
```



Move the player right

Challenge Time

Open the Session 6 starter code: <https://github.com/annisafzhn/CCA.git>

Find the Session 6 folder and run the starter code.

1

Change the Speed

```
player_x += 20
```

Change 20 to:

5

10

50

What changes?

2

Change the Direction

```
player_x -= 20
```

What happens if you change it to:

```
player_x += 20
```

Test → Observe → Explain

3

Add a New Key

Can you make the UP key move the player?

```
pygame.K_UP
```

Which variable should change?

Create Your Own Control

Try changing the movement. Choose:

**KEY**

LEFT

RIGHT

**MOVEMENT**

5

10

20

50

**ACTION**

Move left

Move right

Then :



Today's Big Ideas

EVENT

Something that happens in the game

`pygame.event.get()`

Listens for events

KEYDOWN

A keyboard key was pressed

K_LEFT

Detect the LEFT key

K_RIGHT

Detect the RIGHT key

KEY → TRIGGER → ACTION

NEXT SESSION

Invisible Walls

Module 2 — Session 7

What happens if the player keeps moving?



Next: [Conditionals](#) → [Boundaries](#) → [Keep the Player Inside](#)

See you in Session 7! 🙌